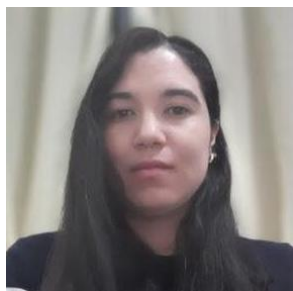


PERSONAL INFORMATION



Yoxara Sánchez Villamizar

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✉ yoxarasv@gmail.com, yoxarasv@sprace.com, y.villamizar@ufabc.edu.br

🌐 <https://yoxara.github.com>

🐙 www.github.com/Yoxara

🚀 www.gitlab.com/yoxara

🆔 **ORCID** [0000-0001-6792-1734](https://orcid.org/0000-0001-6792-1734)

Gender Female | Nationality Venezuelan

FORMAL EDUCATION/DEGREE

April 2023 – April 2025

Post-doc

Federal University of ABC, UFABC, Brazil

The São Paulo Research and Analysis Center, Brazil

August 2019 – February 2023

PhD in physics

Department of Theoretical and Experimental Physics - DFTE.

Federal University of Rio Grande do Norte, UFRN, Brazil

Advisor: Farinaldo Queiroz.

Grantee of: Coordination for the Improvement of Higher Education Personnel, CAPES, Brazil .

August 2018 – August 2019

Master in Physics

Department of Theoretical and Experimental Physics - DFTE.

Federal University of Rio Grande do Norte, UFRN, Brazil

Dissertation title: Phenomenology of the type-II seesaw mechanism.

Advisor: Farinaldo Queiroz.

Grantee of: Coordination for the Improvement of Higher Education Personnel, CAPES, Brazil.

March 2010 – June 2016

Bachelor's degree in Physics

Faculty of Science

University of the Andes. La Hechicera-Mérida, Venezuela.

Thesis title: Optical Properties of ZnO with Mn at 2%.

Advisors:

– Jesús Marquina Castellanos

– Juan Manuel Martín García

September 2002– July 2008

Medium technician in agricultural and animal production.

Boarding school: Instituto San Javier del Valle Grande. El valle - Mérida, Venezuela.

Fe y Alegría

COMPLEMENTARY EDUCATION

September 12 - 23, 2022

Joint ICTP-SAIFR/MITP Summer School on Particle Physics beyond the Standard Model

International Centre for Theoretical Physics, ICTP-SAIFR, São Paulo, Brazil

Credit Hours: 80h

July 4 — July 15, 2022	Summer School on Cosmology 2022 (smr 3720) International Centre for Theoretical Physics, ICTP-Trieste, Italy Credit Hours: 80h
September 5—September 16, 2021	14th MCnet Summer School MCnet
August 23—September 4, 2021	Hadron Collider Physics (HCP) Summer School CERN-Fermilab Credit Hours: 90h
August 2—August 13, 2021	International Neutrino Summer School 2021 CERN European Organization for Nuclear Research Credit Hours: 50h
May 21 — July 11, 2021	Summer School on Particle Physics (smr 3560) International Centre for Theoretical Physics, ICTP-Trieste, Italy Credit Hours: 80h
February 01 – February 05, 2021	Jorge André Swieca Summer School in Particles and Fields Brazilian Physical Society, Brazil Credit Hours: 30 h
September 2020 – January 2021	Special Topics: Introduction to data analysis in High Energy Physics and associated tools - parts I, II and III Armando Dias Tavares Institute of Physics, Rio de Janeiro State University, UERJ, Brazil Credit Hours: 90 h
January 25 – January 29, 2021	Dark Matter School International Institute of Physics, UFRN, Brazil Credit Hours: 11 h
October 12 – October 16, 2020	PASCCO (Particles, Astroparticles, Fields and Cosmology) International Institute of Physics, UFRN, Brazil Credit Hours: 36 h
June 22 — July 3, 2020	II Joint ICTP-Trieste/ICTP-SAIRF School on Particle Physics International Centre for Theoretical Physics - South American, ICTP-SAIRF, Brazil Credit Hours: 56h
January 14 – January 31, 2020	XVII Summer Formative Activities, 2020 Edition, Global Teacher Centro Brasileiro de Pesquisas Físicas, CBPF, Brazil. Credit Hours: 44h
August 2018 – December 2020	English. Education First, EF, Brazil.

2017 – 2018 English.
Cultural Norte Americano, CNA, Brazil.

August – December 2017 Experimental Methods and Techniques in Nuclear and Particle Physics
Institute of Physics, University of São Paulo, , Brazil.
Credit Hours: 180h

2015 To Schedule! An introduction to programming.
Online course, Coursera.org, University of Uruguay and the University of Edinburgh, COURSE-ERA, United States.

2015 Preparing for the AP Physics 1 Exam.
Online course, Coursera.org, University of Houston, COURSEERA, United States.

July 31st, 2008 Communication course: basic and middle levels
Venezuelan Association of the Citizen Band of Mérida. El valle - Mérida, Venezuela.

SCIENTIFIC JOURNAL REFEREE

2020 Journal: Physical Review. D, Particles, Fields, Gravitation, and Cosmology

AWARDS AND TITLES

2021 Awarded in the scientific poster competition at the 28th International Workshop on Weak Interactions and Neutrinos 2021, University of Minnesota

2016 Honors Cum-Laude, University of the Andes.

2008 Recognition for being in the first place in the XXIV promotion of Medium Technicians in agricultural and animal production, U. E. T. San Javier del Valle Grande (Fe y Alegría).

PERSONAL SKILLS

Mother tongue Spanish

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
Portuguese	C1	C1	C1	C1	C1
English	B1	B2	B1	B2	B2
German	A1	A1	A1	A1	A1

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user

Communication skills – **Teamwork:** I have worked in various types of research teams. During my graduation, I worked in experimental physics in material sciences for one year. Also, in 2017 I participated in an experimental research team of nuclear reactions with an interest in astrophysics and exotic nucleus at the Department of Nuclear Physics of the University of São Paulo. During my master's, and PhD I worked in the theoretical research group on astroparticles and particle physics.

Computer skills – competent with latex
– experience with Mathematica and python
– experience with SARAH, FeynRules, CalcHEP, LanHEP and MadGraph5 packages

ADDITIONAL INFORMATION

Events Organisation Committee

Mini-course: Indirect Dark Matter Detection (2022) - UFRN
 Speaker: Farinaldo Queiroz.
 Organizers: Yoxara Sanchez Villamizar and Deivid Silva

Presentations in events

- September 26 - 29, 2022 Yoxara. S. Villamizar . "Implications of the Muon Anomalous Magnetic Moment and the NA62 result for 3-3-1 Models". 2022. **Oral presentation** - Brazilian Physical Society Spring Meeting - II EPSBF 2022.
- September 12 - 23, 2022 Yoxara. S. Villamizar . "Constraining 3-3-1 Models at the LHC and Future Hadron Colliders". 2022. **Student talk** - Joint ICTP-SAIFR/MITP Summer School on Particle Physics beyond the Standard Model.
- July 6 – 13, 2022 Yoxara. S. Villamizar . Constraining 3-3-1 Models at the LHC and Future Hadron Colliders . 2022. **Oral presentation** - International Conference on High Energy Physics (ICHEP) 2022.
- May 16 – 21, 2022 Yoxara. S. Villamizar . Constraining 3-3-1 Models at the LHC and Future Hadron Colliders . 2022. **Poster presentation** - Tenth Annual Large Hadron Collider Physics (LHCP2022).
- July 12 – 14, 2021 Yoxara. S. Villamizar. Dead or Alive? Implications of the Muon Anomalous Magnetic Moment for 3-3-1 Models. 2021.**Oral presentation**.-2021 Meeting of the Division of Particles and Fields of the American Physical Society (DPF21)
- 10-12 November 2021 Yoxara. S. Villamizar. Implications of g-2 for 3-3-1 models, 2021.**Oral presentation**.-ANOMALIES 2021 INTERNATIONAL CONFERENCE
- July 26 – 30, 2021 Yoxara. S. Villamizar. Implications of the Muon Anomalous Magnetic Moment for 3-3-1 Models 2021.**Oral presentation**.-The European Physical Society Conference on High Energy Physics (EPS-HEP)
- November 12, 2021 Yoxara. S. Villamizar. Rare Kaon Decay to Missing Energy: Implications of the NA62 Result for a Z' Model. 2021.**Oral presentation**.-Graviticulas.
- July 27, 2021 Yoxara. S. Villamizar. Implications of g-2 for 3-3-1 model, 2021. **Poster presentation**.-ISAPP 2021 Valencia – Neutrino Physics, Astrophysics and Cosmology
- June 7-12, 2021 Yoxara. S. Villamizar. Implications of the Muon Anomalous Magnetic Moment for 3-3-1 Models. 2021**Poster presentation**.-Weak Interactions and Neutrinos
- Aug 23 – 28, 2021 Yoxara. S. Villamizar. Implications of the Muon Anomalous Magnetic Moment for 3-3-1 Models. 2021**Oral presentation**.-The XXVIII International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2021)
- Jun 14 – 18, 2021 Yoxara. S. Villamizar. Implications of the Muon Anomalous Magnetic Moment for 3-3-1 Models. 2021**Oral presentation**.-PASCOS 2021 the 26th International Symposium on Particle Physics, String Theory, and Cosmology.
- February 01 – 05, 2021 Yoxara. S. Villamizar. Dead or Alive? Implications of the Muon Anomalous Magnetic Moment for 3-3-1 Models. 2021 .**Poster presentation**.-Jorge André Swieca Summer School XXI in Particles and Fields
- January 14 – 21, 2021 Yoxara. S. Villamizar . Dead or Alive? Implications of the Muon Anomalous Magnetic Moment for 3-3-1 Models. 2021. **Oral presentation** - Conference: IAS Program on High Energy Physics (HEP).
- September 02 – 13, 2019 Yoxara. S. Villamizar . Phenomenology of the Type-II seesaw mechanism. 2019. **Poster presentation**- DARK MATTER AND WEAK INTERACTIONS (DARKWIN) CONFERENCE.
- December 04 – 05, 2019 Yoxara. S. Villamizar . Phenomenology of the Type-II seesaw mechanism. 2019. **Oral presentation**- Week of the Post-Graduate Program in Physics of the Federal University of Rio Grande do Norte.
- November 2014 Yoxara. S. Villamizar. XIV Meeting with Physics, Chemistry, Mathematics and Biology. Presenting the experiment: Decomposition of light through a diffraction grating and a prism. (Participation In Events/ **Oral presentation**).
- November, 2013 VII Simposio Latinoamericano de Ingeniería de Superficie. 2013. (Participation In Events/ Seminary).

Evaluation of conference presentations

- November, 2021 XXXII Scientific and Technological Initiation Congress of UFRN 2021. Discussion about work: A Practical Introduction to Knots Invariants (ET1192). Author: Luigy Madson Santana Pinto. Advisor: Dmitry Melnikov
- November, 2021 XXXII Scientific and Technological Initiation Congress of UFRN 2021. Discussion about work: Quantum correlations and AdS/CFT correspondence (ET1217). Author: Joao Victor Ferreira Lacerda Aires. Advisor: Dmitry Melnikov
- November, 2021 XXXII Scientific and Technological Initiation Congress of UFRN 2021. Discussion about work: Applications of the correspondence between theories of modified gravity and General Relativity (ET0720). Author: Erick Augusto Bezerra Alves. Advisor: Emanuele Orazi

Research Projects

- 2019-Present **Finding and building new solutions Beyond Standard Model with extensions from the fermionic and vectorial sector.** *Description:* We will build and explore models that seek to explain dark matter, provide neutrinos with mass, and analyze the muon anomalous magnetic moment, lepton flavor violation, and collider physics, among other problems. In this sense, we will use computational tools used in high-energy particle physics, such as Mathematica, MadGraph5, SARAH, Feynrules, and CalcHep, to build and analyze the models phenomenologically.
- 2018-2019 **Phenomenology of the type-II seesaw mechanism.** *Description:* We reviewed the theoretical and experimental aspects of the type-II seesaw model. This model is a mechanism that generates neutrino masses by adding a weak-scalar triplet to the Standard Model spectrum. This model is phenomenologically attractive because it can generate neutrino masses without invoking new physics effects at high energy scales. We revisited the constraints coming from the ρ -parameter and electroweak precision data, and we discussed collider signatures of the type II seesaw model focusing on the Large Hadron Collider. We exhibited the type-II seesaw model's viable parameter space and showed the complementarity between collider physics and lepton flavor violation. The project represented the first step toward our goal: to assess the High-Luminosity and High Energy LHC sensitivity to the type II seesaw model connected to lepton flavor violation and neutrino mass ordering (e-Print: 1909.07429 [hep-ph]).
- 2015-2016 **Optical Properties of ZnO with Mn at 2%.** *Description:* Mn-doped ZnO ceramic pellets were sintered at two temperatures of 500°C and 900°C. The materials' characterization was carried out by X-ray diffraction, scanning electron microscopy, optical absorption, photoluminescence, and FT-IR techniques.

Additional projects

- 2021 **EPA: ENSINO DE PARTÍCULAS E ASTROPARTÍCULAS (PARTICLE AND ASTROPARTICLE TEACHING).** *Description:* This project aims to divulge Basic Science! Specifically the areas of Particle and Astroparticle Physics. The International Institute of Physics (IIF) of the Federal University of Rio Grande do Norte (UFRN) will be the pole of this gigantic project that will cover several countries, such as the United States, Canada, Colombia, and Chile. Among the activities to be developed, we highlight the production of ebooks, posters, and educational videos aimed at scientific dissemination. In addition, we will offer continuing education courses for teachers and students of Basic Education, who are the main elements of our target audience. With this, we intend to play an important role in training and motivating the new generation.

Bibliographical Production

Thesis and Dissertation

- 1 Yoxara Sánchez Villamizar. Physics beyond the standard model using 3-3-1 models. Ph.D. thesis, Rio Grande do Norte U., 2023. Available: <https://repositorio.ufrn.br/handle/123456789/52222>.
- 2 Yoxara Sánchez Villamizar. Phenomenology of the type-II seesaw mechanism. Master's thesis, Rio Grande do Norte U., 2019. Available: <https://repositorio.ufrn.br/handle/123456789/28261>.
- 3 Yoxara Sánchez Villamizar. Optical Properties of ZnO with Mn at 2%. Bachelor's thesis, 2016. Available: http://platon.serbi.ula.ve/librum/librum_ula/ver.php?ndoc=303043.

Conference proceedings

- 1 Yoxara S.Villamizar, Farinaldo S. Queiroz, Álvaro S. de Jesus, Sergey Kovalenko, C.A. de S. Pires, Implications of the Muon Anomalous Magnetic Moment for 3-3-1 Models. Online conference. Proceedings of The European Physical Society Conference on High Energy Physics. PoS(EPS-HEP2021). Trieste: Sissa Medialab, 2021. p. 638.

Papers in Journals and arXiv

INSPIRE-HEP